SARA (Smart Assistive Reading Aid)

An Offline Smart Glass for Visually Impaired People that Reads Bangla and English Text, Detects Objects, and Recognises and Checks Bangladeshi Taka Notes

All features of the project · 12 October 2026

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1. What SARA is

SARA is a smart glass for blind and low-vision people in Bangladesh. A camera on a spectacle frame, a Raspberry Pi 5, five push-buttons and an earphone let the user read printed Bangla and English, hear what objects are in front, know which Taka note is in the hand, and get careful advice on whether the note is genuine. It speaks Bangla, and everything except one optional mode works without the internet.



The prototype worn during a trial session.

2. Features at a glance

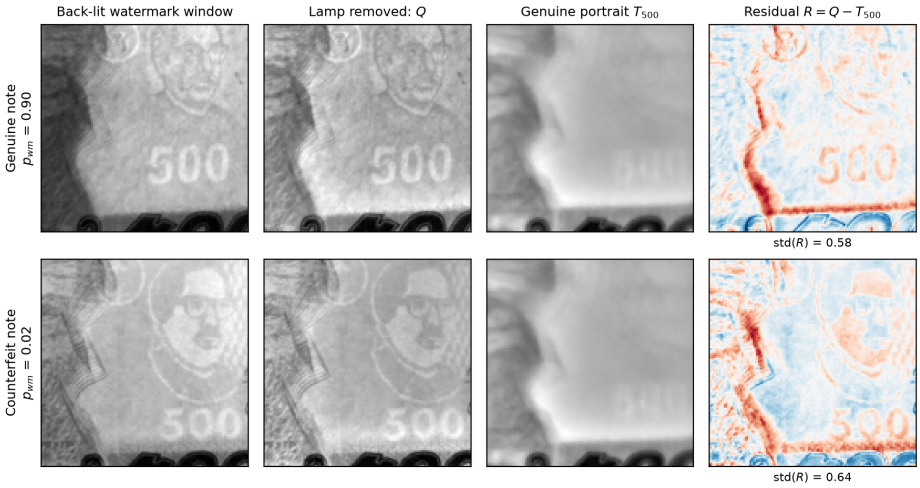
Table 1: The modes of the glass

Feature	What the user gets	How it works	Measured result
Text reading	Printed Bangla and English read aloud; READ repeats it; whole pages are spoken line by line	EasyOCR, with only the main block of text kept, tuned settings and clean-up rules	Letter error 9.3% → 2.4% (Bangla 11.9% → 3.2%) on the test set
Object announcement	Names of objects in front, in Bangla, every 1.5 s	YOLOv8 with a Bangla name for each of 80 objects	mAP@0.5 0.604 on 250 test photos; 207 ms on the Pi 5
Taka recognition	Which of the 9 notes (2 to 1,000 Taka) is in view, with a short vibration	YOLOv8s trained on 22,333 automatically labelled pictures	mAP@0.5 0.995 on held-out pictures; 91.5% on 1,536 photos by other people
Safe counterfeit check (500, 1,000 Taka)	"Likely genuine" or "check by hand"; never "counterfeit"	Multi-view network (PRMVT) with a threshold fixed on validation notes	0 of 88 test counterfeits passed as genuine

Feature	What the user gets	How it works	Measured result
Watermark check	Spoken guidance to hold the note against a light, then "watermark clear" or "check by hand"	Learned watermark finder + MobileNetV2 classifier	92.6% on unseen counterfeit prints; 50 ms on the Pi 5
UV fibre check (200, 500, 1,000 Taka) — new	"UV marks found, likely genuine" or "UV marks not found, check by hand"	Counts the glowing security fibres under a UV lamp (Section 4)	Prototype: see Section 4
Emotion	The facial expression of the person in front	YuNet face finder + EfficientNet-V2-S	86.5% on the RAF-DB test set
Online description (optional)	A spoken description of the scene	A cloud vision model; needs the internet and a key	Not part of the offline results

3. New methods developed in this project

- **Training pictures without hand labelling.** 5,073 note photos are pasted onto real photographs with random angle, size, light and blur, and the program writes each box label itself: 22,333 labelled pictures, no manual work.
- **Prefix training (PRMVT).** A counterfeit model trained on six views fails when the user captures fewer. Training on a random number of views (1 to 6) keeps it accurate: 98.2% with one view against 71.2% (three training runs).
- **Print-disjoint test.** We found that most counterfeit notes in the public dataset are copies of a few prints (279 of 322 readable 500 Taka counterfeits carry one serial number), and built a test in which every counterfeit print is new. On it, adding the watermark check raises accuracy from about 90% to 95.0%.
- **DMWR watermark method.** The lamp is removed from the back-lit photo (Q = I / (G□I)), the watermark is compared with a "genuine portrait" of the note value built from genuine notes only, and what is left over is classified. With 47,818 parameters (0.21 MB) it reaches 91.5% on unseen prints, statistically level with a MobileNetV2 that is 46 times larger.
- **Safe spoken verdict.** The threshold is the highest score any validation counterfeit received, so a new counterfeit passes with probability 1/(n+1) = 1.1%. The glass never accuses.
- **UV fibre check (ENFR).** Described next.

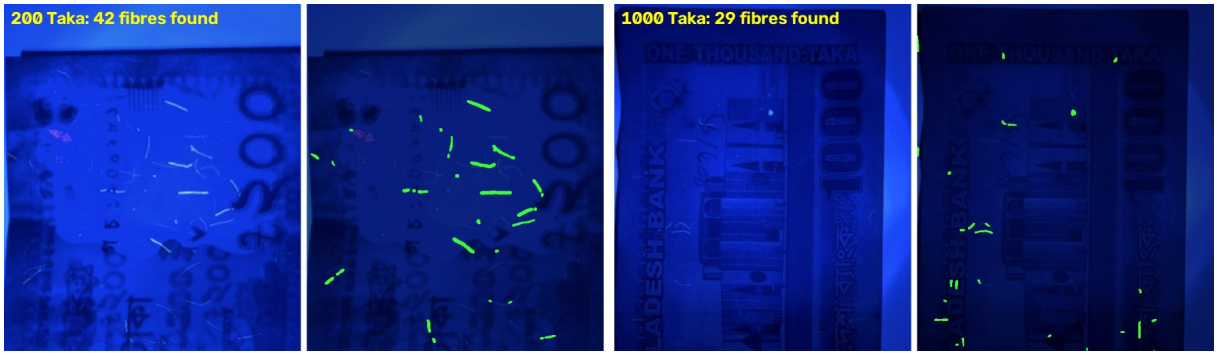


DMWR on a genuine and a counterfeit 500 Taka test note: photo, lamp removed, genuine portrait, residual.

4. New: UV security-fibre check

What the UV light shows. Our own photographs of genuine 200, 500 and 1,000 Taka notes under a UV lamp show short fibres that glow, on paper that stays dull, while the plain white surface beside the note glows as a whole. So a genuine note *has* UV marks; a note without them, or one whose paper glows, is the suspicious one.

Method (excitation-normalised fluorescence residual, ENFR). The lamp's own light reaches the camera as blue. Anything that fluoresces gives off another colour. For the paper of one note, green and red are a fixed share of blue (a = median G/B, b = median R/B). What that share does not explain is fluorescence: E = max(G − a·B, R − b·B). Thin bright structures in E, measured relative to the local lamp level, are the fibres; the check counts those that are thin and elongated inside the note box. Because a, b and the lamp level are measured in the same picture, the brightness of the lamp, its distance and the camera's colour balance cancel out.



Genuine notes under UV (left of each pair) and the fibres found by the check (green, right).

Table 2: UV fibre check on 118 phone photographs of genuine notes (threshold fixed on the first 60%, tested on the last 40%)

What was measured	Result
Photographs / note box found by the detector	118 / 111
Genuine test notes accepted ("UV marks found")	29 of 43 (67%)
200 Taka / 500 Taka / 1,000 Taka	19 of 19 / 7 of 16 / 3 of 8
Plain-surface patches beside the note rejected (test)	21 of 21

What was measured	Result
Same test photos made half as bright / 1.5 times brighter	67% / 74% accepted (brightness changes little)
Same test photos slightly out of focus / at half resolution	30% / 44% accepted (sharpness matters most)

Reading these numbers. The check works when the photo is close and sharp: every 200 Taka test note passed. The misses are photos taken from far away or of the back of the note, where only a few faint fibres are visible. On the glass, the lamp and camera should sit in a small hood about 10 to 15 cm from the note. Two things are not measured yet: real counterfeit notes (none were available to us) and the glass's own camera. For that reason the check is switched off by default, and it says "check by hand" instead of "counterfeit".

5. Speech and interaction

- Bangla speech through espeak-ng and English through Piper, with the language of each sentence chosen automatically.
- **Fixed on 12 October 2026:** Bangla reading used to break up. Every comma, digit and full stop was cut out as a separate piece with its own pause, long texts lost their first sentences, and each Bangla piece first tried a voice that is known to fail on the Pi. Sentences now stay whole, long texts are spoken complete and in order, and the failing voice is tried once per run.
- Five buttons found by touch: MODE, ACTION, READ, VOLUME UP, VOLUME DOWN. Every change is spoken; a vibration confirms a note.
- A field log and a spoken questionnaire for trials with visually impaired users.

6. Speed on the Raspberry Pi 5

Table 3: Measured on a Raspberry Pi 5 (8 GB), 100 calls per part

Part	Typical time
Currency answer (find the note + safe check)	327 ms
Watermark check	50 ms
Object scan	207 ms
Emotion	48 ms
Text reading, one picture	18.7 s (measured before the speed-ups below)

Thirty minutes of mixed use stayed within 1.48 GB of memory, peaked at 77.1 °C and never throttled. For pages, a greedy decoder reads a 12-line page 27% faster with the same accuracy, and lines are spoken as soon as they are read (measured on a laptop limited to four cores; still to be timed on the Pi). A faster setting (OCR_CANVAS_SIZE=640) was about three times faster on the Pi at about half a point more error and can be switched on when speed matters more.

7. What is still open

- UV check: photographs through the glass with the lamp in a hood, and tests on real counterfeit notes with a bank or the police.
- Timing the new text-reading and speech changes on the Pi, and tuning the glass camera for small text and notes.
- Completing the trials with visually impaired users that have started.

On a patent. DMWR and the UV fibre method were designed in this project, but UV fibres and quotient images are known ideas in banknote checking. Whether the specific methods can be patented needs a prior-art search by a patent attorney; this document makes no claim about that.